

CIQ-Expand: Congestion-Interaction QUBOs for Quantum-Enabled Distribution Grid Expansion

Phase 1 Concept Proposal · 2026 Global Quantum + AI Challenge · Enterprise problem statement: **E.ON — Quantum-Enabled Grid Expansion Planning for Distribution System Energy Networks**

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Paradigm Hybrid classical–quantum; gate-based variational (QAOA family) sampling with an AI surrogate loop and classical HPC for grid physics
Platform IBM Heron-class superconducting QPU (≤ 156 qubits) · Qiskit-native code with an E.ON re-execution layer (challenge §5.3) · open data

Summary. We propose **CIQ-Expand**, a benchmark-first pipeline for congestion-driven expansion of distribution grids. It has four stages: **(i)** build a library of *certified* classically hard expansion instances, following the QOBLIB methodology [5], by sweeping physically meaningful “hardness knobs” on open MV/LV grids; **(ii)** compile each instance into a *Congestion-Interaction QUBO* whose couplings come exactly from low-rank power-flow updates and capture how candidate lines substitute for, complement, or (through Braess-type loop flows) work against each other; **(iii)** sample it with hardware-aware QAOA on up to 156 qubits; **(iv)** refine it with a factorization-machine surrogate and validate every proposed plan with full AC power flow. The PoC delivers a Qiskit-executable package, a public hard-instance library with best-known solutions, and an honest head-to-head against Gurobi, heuristics and MPS simulation. This is the evidence E.ON needs to judge where quantum optimization can matter for grid planning.

1 Problem Framing

1.1 Our interpretation of the E.ON problem

A DSO must reinforce its medium- and low-voltage (MV/LV) network so that growing demand from heat pumps and EV charging, and growing injection from rooftop PV, can be served within thermal ratings and voltage limits. We model this as choosing a subset of **candidate lines** K to add to an existing grid: new feeder ties, parallel cables, or substation-to-load-centre links. A plan $x \in \{0,1\}^{|K|}$ has an investment cost $\sum_k c_k x_k$ and a **residual congestion** $Q(x)$. $Q(x)$ is the overload that remains, summed over all lines and weighted operating scenarios, after power flows redistribute in the reinforced grid. The goal (challenge §4.1, §5.2) is to find plans that minimise congestion per line at acceptable cost. For each plan we report the lines added, the objective value, and the reduction in *congestion* events (loading $> 100\%$) and *near-congestion* events (loading $> 90\%$).

1.2 Why the problem is classically hard

- **Interacting decisions.** The value of a line depends on which other lines are built. Two ties relieving the same feeder are *substitutes*. Two lines that are only useful when they close a loop together are *complements*.
- **Non-local physics.** Kirchhoff’s laws redistribute flows across the whole grid, so a new line can *increase* loading elsewhere (Braess’s paradox in power grids [15]). Exact MILP models capture this with disjunctive big-M constraints. Their LP relaxations are notoriously weak, so branch-and-bound closes the optimality gap slowly [16].
- **Scenario and topology coupling.** One plan must work across many operating snapshots. Radial operation adds Steiner-tree and reconfiguration structure, which is strongly NP-hard [2, 3]. As the challenge notes, planners therefore evaluate few options and settle for conservative, costlier plans.

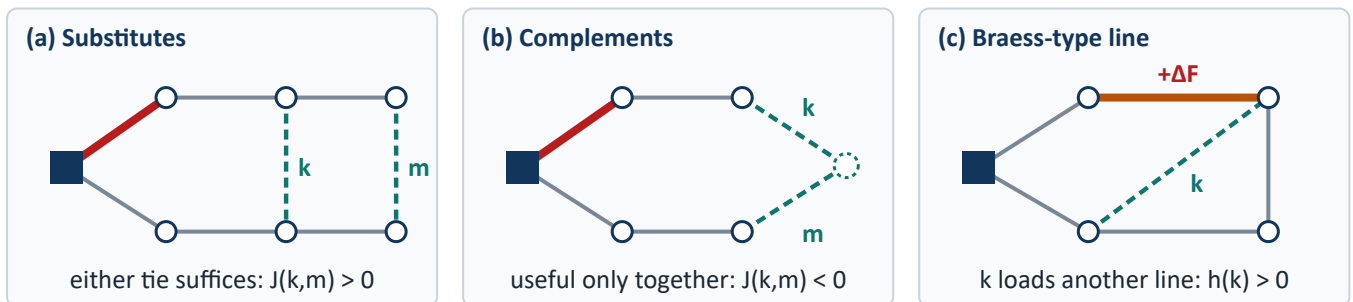


Figure 1. Why candidate lines interact (schematic feeders: ■ substation, thick red = overloaded line, orange = line near its limit, dashed = candidate lines). The CI-QUBO coefficients (§2.2) encode each effect by sign. A mix of all three produces the frustrated energy landscapes that trap classical local search.

1.3 Why quantum methods are a credible lever, and what we will (not) claim

Our key observation is that residual congestion, written as a pseudo-Boolean function of the build decisions, is dominated by low-order interactions between *electrically close* candidates. The optimisation core is therefore a **sparse, frustrated QUBO whose locality is inherited from the grid topology**. This structure suits gate-based QAOA-type sampling for two reasons:

- Short-range couplings map onto native qubit connectivity with shallow circuits.
- Frustration (substitutes vs. complements vs. Braess-type lines) creates rugged landscapes in which classical local search gets trapped.

Evidence of QAOA scaling advantage on a classically hard problem [12] and utility-scale circuits on 100+ qubits [13] justify a rigorous test. We do **not** presuppose advantage. Following [5, 14], we first certify classical hardness, then compare against the best classical solvers at equal wall-clock time, and we report negative results too. A second benefit holds even without speed-up: a sampler returns a *portfolio* of diverse near-optimal plans and, with multi-objective sampling [6], a cost–congestion Pareto front. That is what engineers need, because permitting and constructability decide what finally gets built.

2 Technical Approach

CIQ-Expand is a **hybrid classical–quantum, gate-based variational** workflow in four stages (Fig. 2). Quantum resources are spent only on the combinatorial core. Continuous grid physics stays on classical HPC, and an AI surrogate closes the loop.

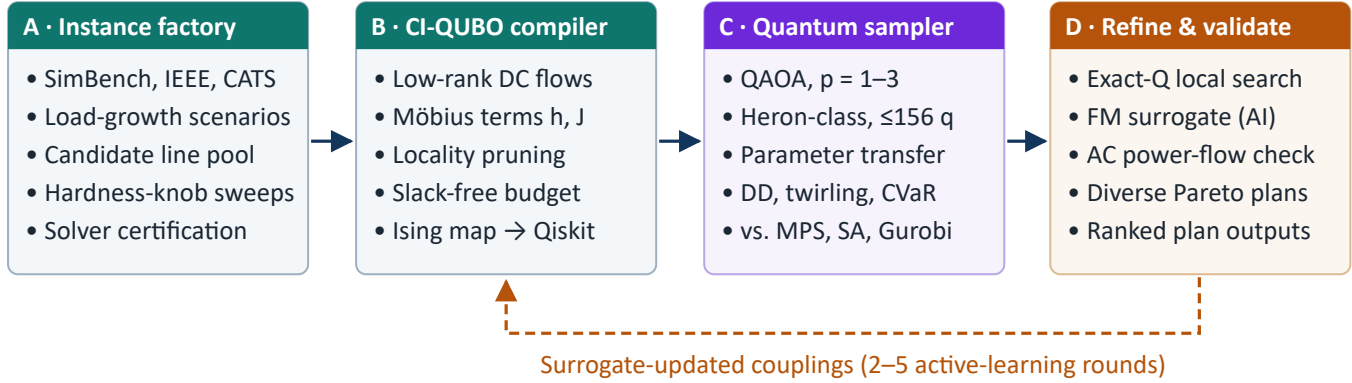


Figure 2. CIQ-Expand pipeline. Header colour shows where each stage runs: teal = classical HPC, violet = quantum processor, amber = AI-driven refinement. The dashed loop feeds couplings learned from exact evaluations back into the QUBO.

2.1 Stage A — Certified hard-instance factory (primary objective)

- **Grids:** SimBench MV/LV networks, which represent German distribution grids where E.ON operates several DSOs [19]; CIGRE MV, IEEE 33/69-bus and pandapower LV feeders [18]; and the CATS and TAMU cases (challenge §6). We also generate synthetic grids that preserve their degree distribution, meshedness and feeder-length statistics, so problem size can be varied smoothly.
- **Scenarios:** SimBench time series with heat-pump, EV and PV growth multipliers, clustered to $|S| = 4-24$ representative snapshots.
- **Candidates:** geographically plausible ties with length- and type-based costs.
- **Hardness knobs:** load growth near the congestion onset, candidate density $|K|/|L|$, share of loop-closing (Braess-prone) candidates, budget tightness and scenario diversity.
- **Two formulations per instance:** an exact multi-scenario MILP (F1, a line-inclusion model in the spirit of [1, 4], with DC or LinDistFlow physics and optional radiality; App. A) and the quantum-native CI-QUBO (F2).
- **Certification:** both forms are solved with Gurobi and CPLEX (HiGHS/SCIP for open reproducibility) at 1 h and 4 h limits with 5 seeds. An instance is **hard** if optimality is not proven within 1 h, or if runtime grows exponentially across its size family. The library is exported in QOBLIB format [5] with best-known solutions (BKS).

2.2 Stage B — Congestion-Interaction QUBO (CI-QUBO)

Let $f_s(x)$ be the DC line flows in scenario s . Residual congestion is $Q(x) = \sum_s \pi_s \sum_\ell [|f_{\ell s}(x)| - F_\ell]^+$. Adding line k is a rank-one update of the nodal susceptance matrix, so flows for any small set of additions follow *exactly* from the base PTDFs via Sherman–Morrison–Woodbury, with no refactorisation. We expand Q in its Möbius (pseudo-Boolean) basis around the existing grid and truncate at second order:

$$\hat{Q}(x) = Q(0) + \sum_k h_k x_k + \sum_{k < m} J_{km} x_k x_m, \quad h_k = Q(e_k) - Q(0), \quad J_{km} = Q(e_k + e_m) - Q(e_k) - Q(e_m) + Q(0)$$

Properties of the expansion:

- **Exact where it matters:** $\hat{Q}$ equals Q for every plan with ≤ 2 additions.
- **Cheap to build:** it needs only $1 + n + n(n-1)/2$ low-rank evaluations per scenario (11,326 for $n = 150$ candidates), an embarrassingly parallel HPC task.
- **Interpretable:** the coefficients generalise the influence scores of [4]. $J_{km} > 0$ marks substitutes, $J_{km} < 0$ complements, and $h_k > 0$ flags Braess-type candidates (Fig. 1).
- **Sparse:** couplings decay with electrical distance, so pruning $|J_{km}| < \varepsilon$ leaves a sparse interaction graph.

The cost Hamiltonian is $H(x) = \sum_k c_k x_k + \rho \hat{Q}(x) + P_B(x)$. The budget term $P_B(\sum c_k x_k \leq B)$ takes one of two forms:

- **Dense tiers: unbalanced penalisation** [9], $P_B = -\lambda_1 g + \lambda_2 g^2$ with $g = B - \sum c_k x_k$. It needs no slack qubits, so **one qubit = one candidate line**.
- **Hardware-native sparse tiers:** a quadratic penalty would add all-to-all couplings, so the budget instead enters as a sparsity-preserving Lagrangian term $\lambda \sum c_k x_k$. Sweeping λ traces the cost–congestion Pareto front, and sample repair guarantees feasibility.

Cardinality budgets (“at most m lines”) can instead be enforced exactly with XY mixers [8]. Interactions among three or more lines are handled in Stage D.

2.3 Stage C — Quantum sampling on NISQ hardware

- **Algorithm:** QAOA [7] at depth $p = 1\text{--}3$ via Qiskit Runtime SamplerV2 on IBM Heron-class processors (heavy-hex, ≤ 156 qubits), with Ising map $x_k = (1 - Z_k)/2$.
- **Hardware-aware compilation:** the electrically local interaction graph is laid out on the heavy-hex lattice. For the largest instances, hardware-native sparsification keeps only couplings realisable without SWAPs, giving CZ depth 6 per QAOA layer. Dropped weak couplings are restored in classical scoring. Denser instances up to about 60 qubits use SWAP-network or Sabre routing.
- **Parameters without costly QPU loops:** angles are optimised by statevector simulation on ≤ 30 -qubit instances and transferred to large ones, exploiting parameter concentration [11, 25]. Short CVaR fine-tuning [10] is applied only where transfer degrades.
- **Error suppression:** dynamical decoupling, Pauli twirling and readout mitigation. We need good *samples*, not precise expectation values, so expensive estimator-level mitigation is avoided.
- **Multi-objective mode:** weighted scalarisations of (cost, congestion, losses) share transferred parameters, as in [6], producing a Pareto front from one batch of circuits.

2.4 Stage D — Classical refinement, AI surrogate and physical validation

- **Refine:** every sample is repaired for budget feasibility, polished by steepest-descent bit-flip local search on the *exact* $Q(x)$, and scored.
- **Learn:** the elite plans train a **factorization-machine surrogate** [17], which has QUBO form. Its fitted coefficients update the CI-QUBO for the next round. This quantum–AI active-learning loop (2–5 rounds) captures higher-order interactions.
- **Validate:** final plans are checked with **AC power flow in pandapower** across all scenarios: thermal loading, EN 50160 voltage band 0.9–1.1 p.u., losses, and N-1 for MV.
- **Output:** lines added, cost, congestion and near-congestion reduction, and a ranked, diverse Pareto set.

3 Feasibility and Resource Requirements

Resource	Requirement (16-week PoC)	Notes and assumptions
Quantum hardware	IBM Heron-class QPU (133–156 qubits); $\approx 10\text{--}15$ QPU-hours (≈ 250 sampler jobs of $10^4\text{--}10^5$ shots)	Via Challenge compute partner / IBM Quantum. Parameter transfer keeps variational QPU loops short. Tier budget in App. B.
Classical compute	1 CPU node (64 cores, 256 GB) + 1–2 data-centre GPUs	Solver baselines, low-rank power flows, local search; GPU statevector (≤ 32 q) and MPS (bond dim. $\chi \leq 1024$) in Qiskit Aer.
Software	Python, Qiskit 2.x, qiskit-ibm-runtime, qiskit-aer, pandapower, SimBench, NetworkX, PyTorch; Gurobi (academic), CPLEX, HiGHS/SCIP	Open source throughout, except commercial MILP solvers, which serve only as baselines.
Data	Open grids and time series (challenge §6, SimBench); E.ON anonymised MV/LV subgrids when released	Loaders ingest pandapower / MATPOWER / CIM, so E.ON data needs no code change.
People	Three-person team covering quantum optimisation, AI/ML orchestration and software engineering	Power-systems advisory input sought at UPM (§7).

Circuit budget (largest tier). A 156-qubit heavy-hex device has ≈ 176 couplers. A hardware-native CI-QUBO layer therefore needs ≤ 352 CZ gates at CZ depth 6, or ≈ 700 CZ at $p = 2$. At today’s two-qubit error rates the overall circuit fidelity will be low but not negligible. Our design relies on *sample quality after local search* rather than noiseless expectation values, and the ablation study (M4) will measure whether a quantum signal survives.

Runtime (challenge §5.3). Coefficient generation takes minutes on 64 cores. QPU execution takes under 1 h per instance, excluding queueing. Local search and AC validation take minutes. Target: ≤ 4 h end to end.

Assumptions.

- DC sensitivities rank congestion relief adequately in MV grids. LV feeders with high R/X use LinDistFlow sensitivities [24], and every final plan is AC-validated.
- A second-order surrogate plus FM refinement is accurate enough; this is tested at gate M2.
- QPU access is available in weeks 9–14.

Key risks (full register with likelihood and impact in App. C):

Risk	Mitigation
Noise erases the quantum signal at 100+ qubits	Hardware-native sparsification, $p \leq 2$, error suppression; ablation M4 measures the effect; fall back to the 48–100-qubit tier
Second-order surrogate misses higher-order interactions	Exact re-scoring of every sample, FM active learning, optional 3-local terms; go/no-go gate M2
Instances not hard for Gurobi at NISQ-compatible sizes	Systematic knob sweeps, exact-model (F1) hardness, QOBLIB-style “hard-for-size” reporting
Shallow circuits turn out to be classically simulable	Explicit MPS and light-cone study (M5); move to $p \geq 2$; claim only what survives
Limited in-house power-systems background	Validated open tools (pandapower, SimBench), automatic AC checks, UPM advisory input and early E.ON review of assumptions (§7)

4 Expected Impact

A successful PoC changes how expansion options are generated, compared and justified:

Planning aspect	Current practice (challenge §3)	With CIQ-Expand
Option generation	Rules and experience; a few candidates iterated through simulation	Systematic candidate pools; thousands of plans physics-scored automatically
Line interactions	Assessed implicitly, one option at a time	Explicit h/J sensitivities flag substitutes, complements and Braess-type lines
Scenario coverage	Limited by spreadsheet and engineer effort, leading to conservative plans	All scenarios scored jointly for every candidate plan
Planner output	One recommended plan	Ranked, diverse Pareto set with cost, congestion, losses and AC checks
Quantum evidence	Anecdotal or vendor claims	Certified-hard instances, ablations and MPS crossover; open and reproducible

- **For E.ON planning teams.** Planners move from a handful of manually screened options to thousands of physics-scored plans and an explicit cost–congestion trade-off curve. The h/J sensitivities expose counter-intuitive interactions that drive over-investment, such as redundant ties and Braess-type lines, extending the influence-score practice of [4]. **PoC targets** on open and anonymised benchmark grids:
 - **≥ 90% reduction in congestion events** versus the unreinforced grid.
 - **Investment cost within 5%** of the best classical plan.
 - **≥ 10 distinct plans within 2%** of the best-known objective per instance, for engineer-in-the-loop selection.
- **For quantum benchmarking.** The first open, certified-hard, physics-grounded grid-expansion instance library in QOBLIB format with BKS. It comes with a transparent comparison of QAOA sampling against Gurobi, simulated annealing, tabu search and MPS simulation at 100–156 qubits, published whether the outcome is positive or negative.
- **Path to utility scale.** Qubit count grows linearly with the candidate pool. Regional planning areas with several hundred candidates can be decomposed by substation service area, and the pipeline carries over unchanged to larger future QPUs. We aim for a joint scientific publication with E.ON (challenge §5.2).

Scalability path (challenge §4.2: from a few qubits to utility scale, > 100 qubits):

Stage	Grid scope	Candidate lines = qubits	Execution mode
PoC-S	Single LV/MV feeder	16–32	Statevector and QPU; exact optimum known for validation
PoC-M	MV feeder group	48–100	QPU with routing; scaling curves and ablations
PoC-L	MV substation service area	127–156	Heron-class QPU with hardware-native couplings (utility scale)
Roll-out	Regional planning area	300–1,000+	Decomposition by service area today; direct execution on larger future QPUs

5 Validation Plan

ID	Metric	Method	Success criterion
M1	Instance hardness	Gurobi/CPLEX runtime and gap, 5 seeds, 1 h and 4 h limits; scaling fits per size family	≥ 10 certified-hard instances in ≥ 3 size classes, ≥ 3 of them with 100–156 variables

ID	Metric	Method	Success criterion
M2	Surrogate fidelity (gate, wk 8)	Spearman ρ between $\hat{Q}$ and exact Q on 10^4 random and elite plans; DC vs. AC ranking	$\rho \geq 0.9$ and top-10 overlap $\geq 80\%$; otherwise add 3rd-order (HUBO) terms
M3	Solution quality	Gap to proven optimum (exhaustive for $n \leq 32$) or BKS	$\leq 1\%$ ($n \leq 32$), $\leq 3\%$ ($n \geq 100$). <i>Stretch</i> : match or beat the Gurobi incumbent at equal wall-clock on ≥ 1 hard instance
M4	Quantum contribution	Ablation: identical pipeline fed by QAOA samples vs. uniform random samples, SA and tabu at equal evaluation budget	Significant (Mann–Whitney $p < 0.05$) gain in median residual energy or time-to-target on ≥ 1 instance class; reported either way
M5	Classical simulability (secondary obj.)	Qiskit Aer MPS with $\chi \in \{16, \dots, 1024\}$ [20, 26]; light-cone analysis for $p = 1$	Locate the circuit sizes/depths where MPS at equal runtime gives a worse approximation ratio than hardware
M6	Grid outcome	AC power flow on all scenarios (pandapower)	$\geq 90\%$ fewer congestion events; no new voltage violations; loss change reported
M7	Engineering	End-to-end runtime; E.ON re-execution in Qiskit	≤ 4 h per instance (excl. queue); one-command Docker reproduction

Phase 2 plan (16 weeks; Gantt chart in App. C).

- **WP1** Instance factory and hardness certification (wk 1–6).
- **WP2** CI-QUBO compiler and surrogate (wk 3–9), with go/no-go gate M2 at wk 8. Fallback if M2 fails: Benders decomposition of F1 with a QUBO master problem.
- **WP3** Simulation (wk 5–10), then hardware runs (wk 9–14).
- **WP4** Benchmarking, MPS and AC validation (wk 10–15).
- **WP5** Qiskit package, report and paper draft (wk 13–16).

Phase 2 deliverables.

- **D1** Open hard-instance library with best-known solutions (QOBLIB format).
- **D2** Qiskit-native code, Docker image and an E.ON re-execution guide.
- **D3** Benchmark report covering M1–M7, including negative results.
- **D4** Plan reports for E.ON anonymised subgrids: lines added, cost, congestion before and after, AC checks.
- **D5** Joint paper draft.

6 Hybrid and Cross-Domain Integration

Layer	Role in CIQ-Expand	Why it runs there
Classical HPC	Scenario generation; low-rank power flows for h, j ; MILP baselines; local search; AC validation	Continuous physics and linear algebra are solved exactly and cheaply on classical hardware
AI / ML	Factorization-machine surrogate (active learning); hardness predictor over instance features [23] that steers generation toward hard regions; QAOA angle prediction from graph features	Learns high-order structure from data and cuts the number of QPU calls
Quantum	Sampling the sparse, frustrated combinatorial core (CI-QUBO)	The only step where classical heuristics are expected to get trapped; can be switched off for clean attribution (M4)

Fit with DSO workflows. Inputs are the network models and scenario sets that planners already maintain. Outputs are ranked plans in standard formats (pandapower JSON, MATPOWER, CSV) that can be imported into existing grid-analysis tools, keeping the engineer in the loop.

7 Team Capability

Member	Role on this project	Relevant expertise
Kyoung-Tae (Elijah) Kim MSc Quantum Computing, UPM (Spain)	Quantum Project Manager: sprint plan and gates (M2 go/no-go); benchmark harness for instance-knob sweeps and ablation arms (M1, M4); surrogate and hardness-predictor loop (§6); experiment tracking and timing methodology	AI & ML, RAG, LangChain, agentic-AI orchestration and system optimisation
Naveed Qureshi MSc Quantum Computing, UPM (Spain)	Quantum Technical Lead: CI-QUBO formulation and budget encodings (Stage B); QAOA compilation, parameter transfer and hardware runs (Stage C); circuit and QPU resource estimates (App. B)	Satellite mission-planning optimisation using QUBO; constrained discrete optimisation
Alejandro Valencia Blancas MSc Quantum Computing, UPM (Spain)	Quantum Solutions Architect: software architecture and Qiskit re-execution layer; instance factory and pandapower validation (Stages A, D); classical baselines (Gurobi, SA, tabu); verification tests and reproducible open-source release	Software engineering, JavaScript, database systems, mobile communications, AI; ethics and epistemology of AI

Why this team. Entangled Minds formed on the MSc in Quantum Computing at the Universidad Politécnica de Madrid (UPM). Across group projects in quantum algorithms, variational methods and quantum optimisation, the three of us have worked the way this proposal requires: design hybrid quantum–classical systems, validate in simulation first, and benchmark honestly against strong classical baselines. The core of the E.ON problem matches Naveed’s experience directly. Turning satellite mission planning into a constrained QUBO is the same task as encoding line-build decisions under a budget, including the choice between penalty-based and slack-free constraint encodings. Elijah’s AI-orchestration experience covers the surrogate active-learning loop and the many-arm sweeps behind M1 and M4. Alejandro’s software-engineering depth serves the instance pipeline, the Qiskit compatibility layer and one-command reproducibility.

Execution readiness. Power-systems engineering is not our core background, and we have designed the plan around that. We rely on mature, validated open tools (pandapower, SimBench) rather than custom grid solvers. Every final plan is checked automatically with AC power flow, and small instances have exactly known optima, so correctness is self-checking. Gate M2 tests the surrogate before significant QPU time is spent. We will seek advisory input from power-systems faculty at UPM’s School of Industrial Engineering (ETSII), and we would welcome a review of our candidate-line and scenario assumptions by E.ON planning engineers early in WP1. QPU time will come from the Challenge compute partner, and classical compute from cloud rental or academic allocations.

Appendix — CIQ-Expand

Supplementary material to the Phase 1 Concept Proposal (E.ON challenge) · A. Formulations · B. Benchmark protocol & resources · C. Work plan, risks, software · References

A Mathematical Formulations

A.1 F1 — Exact multi-scenario expansion MILP (reference model and hardness certification)

Sets: buses N , existing lines L , candidate lines K , scenarios S with weights π_s ; line ℓ runs from $o(\ell)$ to $d(\ell)$. **Variables:** $x_k \in \{0,1\}$, flows $f_{\ell s}$, angles θ_{ns} , overloads $u_{\ell s} \geq 0$, load-shedding slack $r_{ns} \geq 0$.

$$\begin{aligned} \min \quad & \sum_{k \in K} c_k x_k + \rho \sum_{s \in S} \pi_s \sum_{\ell \in L} u_{\ell s} + \sigma \sum_{s \in S} \pi_s \sum_{n \in N} r_{ns} \\ \text{s.t.} \quad & \sum_{\ell: d(\ell)=n} f_{\ell s} - \sum_{\ell: o(\ell)=n} f_{\ell s} + g_{ns} + r_{ns} = d_{ns} \quad \forall n \in N, s \in S \quad (\text{nodal balance}) \\ & f_{\ell s} = b_{\ell}(\theta_{o(\ell)s} - \theta_{d(\ell)s}) \quad \forall \ell \in L, s \quad (\text{KVL, existing lines}) \\ & |f_{ks} - b_k(\theta_{o(k)s} - \theta_{d(k)s})| \leq M_k(1 - x_k), \quad |f_{ks}| \leq F_k x_k \quad \forall k \in K, s \quad (\text{disjunctive KVL, candidates}) \\ & |f_{\ell s}| \leq F_{\ell} + u_{\ell s} \quad \forall \ell \in L, s; \quad \sum_k c_k x_k \leq B \quad (\text{thermal limits with overload slack; budget}) \end{aligned}$$

Variants.

- **F1-R** adds line-status binaries and single-commodity-flow radiality constraints, which introduces the Steiner/reconfiguration structure of [2, 3].
- **F1-LV** replaces DC physics with LinDistFlow [24]: $v_j = v_i - 2(r_{ij}P_{ij} + x_{ij}Q_{ij})$, with $0.81 \leq v \leq 1.21$ (squared p.u.).

All big-M values are tightened by bound propagation, so the classical baselines are strong, not straw men.

A.2 F2 — CI-QUBO, Ising Hamiltonian, QAOA and surrogate

Ising form. Substituting $x_k = (1 - Z_k)/2$ into $H(x) = \sum c_k x_k + \rho \hat{Q}(x) + P_B(x)$ [22] gives

$$H_C = \sum_k \alpha_k Z_k + \sum_{k < m} \beta_{km} Z_k Z_m + \text{const}, \quad \alpha_k = -(c_k + \rho h_k)/2 - (\rho/4) \sum_{m \neq k} J_{km}, \quad \beta_{km} = \rho J_{km}/4$$

The unbalanced budget penalty adds $\lambda_2 c_k c_m/2$ to every β_{km} , which makes the coupling graph dense. It is therefore used only in tiers S/M. Tier L uses the Lagrangian form, which modifies only α_k .

QAOA and CVaR. The QAOA state is $|\gamma, \beta\rangle = \prod_{j=1..p} e^{-i\beta_j \sum_k X_k} e^{-i\gamma_j H_C} |+\rangle^{\otimes n}$. When tuning is needed, the objective is CVaR_{α} , the mean of the best α -fraction of sampled energies [10].

Factorization-machine surrogate.

$$\hat{y}(x) = w_0 + \sum_k w_k x_k + \sum_{k < m} \langle v_k, v_m \rangle x_k x_m, \quad v_k \in \mathbb{R}^r, r = 8-16$$

It is initialised from (h, J) by truncated eigendecomposition and retrained on the exact Q of all evaluated plans. Because $\hat{y}$ is itself a QUBO, it plugs straight back into Stage C.

B Instance Generation, Benchmark Protocol and Resource Estimates

Hardness knob	Range swept	Mechanism by which it raises difficulty
Load growth α	1.0–2.0, fine grid near congestion onset	Maximises the number of simultaneously interacting congestion events
Candidate density K / L	0.2–1.5	More substitutes/complements $\rightarrow$ denser, more frustrated J
Loop-closing share	0–60% of candidates	Braess-type redistribution $\rightarrow h_k > 0$ and mixed-sign couplings
Budget tightness $B/\sum c$	0.05–0.40	Knapsack-like coupling between otherwise independent feeders
Scenario count S	4–24 snapshots	Multiplies disjunctive constraints in F1; enriches J in F2
Cost heterogeneity	length-proportional ... $\pm 50\%$ noise	Removes trivial greedy (cost-per-relief) orderings

Size classes. $n \in \{16, 24, 32, 48, 64, 96, 128, 156\}$ for QPU-compatible families, plus $\{256, 512, 1024\}$ for classical scaling curves, with ≥ 20 instances per class.

Baselines.

- Gurobi and CPLEX: MILP on F1, MIQP on F2.
- Simulated annealing and tabu search.
- Exhaustive GPU enumeration for $n \leq 32$.

All baselines run with 5 seeds and identical hardware and time logging.

Metrics.

- Optimality gap to the proven optimum or BKS.
- Time-to-target $TTT(\epsilon)$ at $\epsilon \in \{1\%, 3\%\}$.
- Residual-energy distribution of samples and approximation ratio.
- Hypervolume of cost–congestion Pareto fronts.
- Congestion and near-congestion events after AC validation.

Tier	Qubits n	Interaction graph after pruning	QAOA p	CZ / layer	CZ depth / layer	Shots / job	Purpose
S	16–32	avg. degree ≈ 6 ; Sabre routing	1–3	≈ 150 –600	≈ 30 –60	10^4	Exact optimum known; hardware vs. statevector/MPS
M	48–100	degree ≤ 4 ; Sabre routing	1–2	≈ 250 –700	≈ 12 –25	$2 \cdot 10^4$	Scaling curves, ablations (M4)
L	127–156	heavy-hex native, degree ≤ 3	1–2	≤ 352	6	10^5	Hardness regime; MPS crossover (M5)

QPU budget. ≈ 2 h (tier S) + 3 h (M) + 6–8 h (L) + 20% contingency $\approx$ **12–15 QPU-hours**. Gate counts are pre-transpilation estimates and will be confirmed in WP3.

C Work Plan, Risk Register and Software Deliverables

Work package	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Milestone
WP1 Instance factory & hardness																	Library v1 (wk 6)
WP2 CI-QUBO compiler & FM surrogate																	Gate M2 (wk 8)
WP3a Simulation & parameter transfer																	Transferred angles
WP3b Hardware runs (S $\rightarrow$ M $\rightarrow$ L)																	156-q runs (wk 13)
WP4 Benchmarks, MPS, AC validation																	Benchmark report (wk 15)
WP5 Qiskit package, report, paper																	Final delivery (wk 16)

Risk	L / I	Mitigation
Hardware noise erases quantum signal at 100+ qubits	H / M	Hardware-native sparsification, $p \leq 2$, DD + twirling, CVaR; ablation M4 quantifies the effect honestly; fall back to tier M
Second-order surrogate misses high-order interactions	M / H	FM active learning, optional 3-local (HUBO) terms, exact re-scoring of every sample (gate M2)
Instances are not hard for Gurobi at NISQ sizes	M / H	Knob sweeps (Braess share, $ S $), F1/F1-R hardness, QOBLIB-style reporting of “hard-for-size” instances
$p = 1$ circuits are classically simulable (light cones)	M / M	Move to $p \geq 2$ and non-native graphs; M5 characterises this explicitly; claim only what survives
DC approximation inaccurate on LV feeders	M / M	LinDistFlow sensitivities (F1-LV); AC validation of all final plans
QPU access / queue delays; late E.ON data	L / M	Simulator-first, parameter transfer, backend-agnostic Qiskit code; open SimBench grids until data arrive

Software architecture and Qiskit compatibility layer [21]. The repository (Apache-2.0) is organised as follows:

- `ciq.instances`: loaders, generator, hardness certification.
- `ciq.qubo`: low-rank power flow, Möbius coefficients, pruning, penalties $\rightarrow$ `SparsePauliOp`.
- `ciq.solvers`: QAOA via `SamplerV2`, Aer MPS, SA, tabu, MILP.
- `ciq.validate`: AC power flow, grid metrics.
- `ciq.bench`: configs, logging, figures.

Every quantum job is stored as a Qiskit `QuantumCircuit` in QPY together with its transpilation settings. E.ON can re-run any experiment with a single command, `ciq run --instance <id> --backend <ibm_backend | aer_mps>`, inside the provided Docker image. **Deliverables:** code, instance library with BKS, benchmark report, and a draft joint publication.

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